

Separable NM-ROM

Multiresolution results

Poisson 2D · Heat 2D · Burgers 2D · Reflective waves

Tunable NM-ROM project

11 September 2026

Frozen trained weights across all tested resolutions

Paired GPU measurements and independently audited fields. Development cohorts; independent final paper validation remains open.

At 1024 intervals: where the evidence stands

Fastest tested iterative FOM meeting the declared target, divided by ROM time.

Problem	FOM/ROM	ROM error %	ROM status
Poisson 2D	33.91	6.802	Accuracy target missed
Heat 2D	4.85	4.555	Target and stopping pass
Burgers 2D	1.63	3.908	Accuracy passes; stalls*
Reflective waves	0.23	3.568	Full-state target missed

Takeaway

Heat has the clearest qualified speed advantage. Burgers has a smaller fine-grid advantage under its existing stopping contract.

Ratio above one favors the ROM. These are named iterative controls, not the fastest possible FOMs. Errors are worst cases under the panel-specific norms.

*Burgers accepts small-step/improvement stalls; stationarity was not measured. Reflective-wave error here is displacement only.

The measurement contract

- Meshes: $N = 64, 256, 1024$ intervals per axis. One frozen checkpoint per PDE/boundary.
- Online GPU time includes input projection/fitting, the solve or rollout, and requested full fields.
- Training, mesh assembly and compilation are offline. Host transfers are reported separately.
- Each ROM/FOM comparison uses one GPU allocation; costs and errors come from the same invocation.

Problem	Cases	Repeats	Error normalization
Poisson 2D	30	3	Current solution norm
Heat 2D	12	3	Current field norm
Burgers 2D	4	3	Fixed initial field norm
Reflective waves	2	3	Current component norms

Times: medians of all retained repetitions, with equal repeats per case. Errors: maxima over cases, output times and repetitions. All repetitions and outliers are retained.

The target is 5%, with panel-specific refinement and solver checks. Wave qualification includes displacement, velocity and energy state.

Shared decoder; explicitly named full solvers

$$u(x; z) = \sum_{j=1}^R g_j(x) h_j(z), \quad u = G h(z).$$

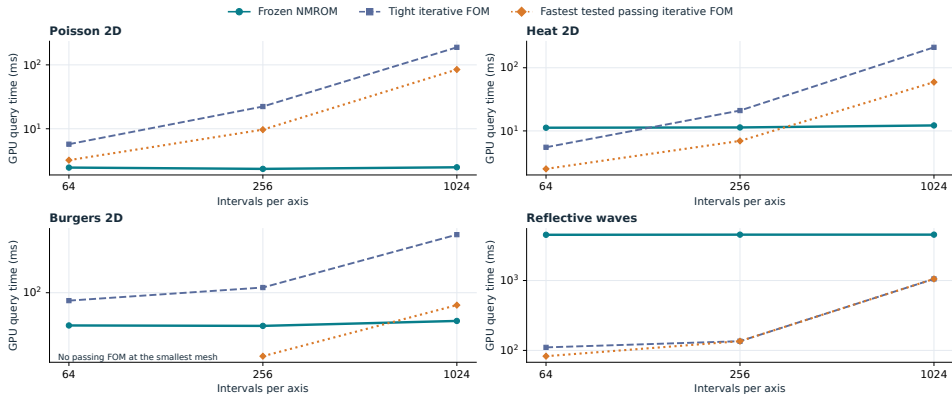
The spatial bank is fixed during a query. The nonlinear coefficient map is evaluated as latent coordinates change.

Problem	Reduced computation	Primary iterative FOM
Poisson	Preassembled weak algebra	CG: 10^{-6}
Heat	Weak CN; Cholesky updates	CN-CG: 10^{-6}
Burgers	Preassembled linear; sampled upwind	Newton 10^{-6} ; linear 10^{-8}
Reflective waves	Nonlinear projected RK4 dynamics	Midpoint CG: 10^{-6}

CG = conjugate gradient; CN = Crank–Nicolson. Burgers uses FFT-preconditioned Newton–BiCGStab. Weak equations average the residual against smooth tests.

Scope: Burgers uses sampled sign-upwind advection, not the historical polynomial tensor. Wave ROM/FOM use the same step size with different integration formulas.

ROM runtime stays nearly flat over these meshes



Logarithmic axes. Orange: fastest tested iterative FOM satisfying the declared target at that mesh. A faster ROM curve alone does not establish an accuracy-qualified win.

This is a finite-range observation. Reading dense inputs and producing full fields still depend on mesh size. Compare methods within each panel, not across different GPU allocations.

Poisson: strong runtime scaling; accuracy needs work

Primary FOM: zero-start, unpreconditioned CG at 10^{-6} .

N	ROM ms	FOM ms	FOM/ROM	ROM error %	FOM error %
64	2.47	5.73	2.32	6.803	0.2572
256	2.36	22.28	9.46	6.802	0.01546
1024	2.49	188.21	75.45	6.802	0.0007351

Takeaway

The decoder transfers across meshes with nearly unchanged error and runtime. Its worst error remains above the 5% target.

At $N = 1024$, loose CG at 10^{-1} takes 84.58 ms with 1.019% error. The raw FOM/ROM ratio is 33.91, but the ROM fails the target.

Error: current-relative solution L_2 , against a refined finite-difference reference. Reference allowance is included in qualification. All recorded ROM solves satisfy stationarity and all CG controls converge.

Heat: a fine-grid speedup that survives looser CG

Primary FOM: matched Crank–Nicolson steps, CG at 10^{-6} .

N	ROM ms	FOM ms	FOM/ROM	ROM error %	FOM error %
64	11.23	5.51	0.490	4.559	0.06738
256	11.34	21.00	1.85	4.556	0.03126
1024	12.21	210.60	17.25	4.555	0.03491

Takeaway

At $N = 1024$, the ROM remains $4.85\times$ faster than the fastest tested passing iterative FOM. Both satisfy the physical target and stopping checks.

Looser CG at 10^{-2} : 59.18 ms and 0.7707% error. Including input/output transfers, the FOM/ROM ratio is 2.27.

Error: current-relative field L_2 , against a refined continuum sine-series solution. Coarse-grid CG remains faster; the medium-grid advantage depends on tolerance.

Burgers: fine-grid gain with stalled solves

Primary FOM: FFT-preconditioned Newton–BiCGStab, $10^{-6}/10^{-8}$.

N	ROM ms	FOM ms	FOM/ROM	ROM error %	FOM error %
64	36.23	78.24	2.16	10.85	10.99
256	35.75	117.47	3.29	4.551	4.027
1024	41.68	605.75	14.53	3.908	2.142

Takeaway

At $N = 1024$: $1.63\times$ faster than the fastest tested passing iterative FOM. The two larger ROM meshes pass the physical target.

Looser FOM: Newton 10^{-2} , linear 5×10^{-1} ; 68.04 ms, 2.39% error. With transfers: $1.44\times$.

Stopping limitation: all ROM fits/steps stop by a small-step/improvement rule; stationarity is unmeasured. Both ROM and FOM miss the physical target on the smallest mesh.

Error uses the fixed initial reference norm. Spatial/temporal reference allowances are included in the physical gate.

Reflective waves: displacement alone is incomplete

Primary FOM: implicit-midpoint CG at 10^{-6} .

N	ROM ms	FOM ms	FOM/ROM	ROM error %	FOM error %
64	4510.12	110.53	0.025	3.641	0.3521
256	4530.34	135.46	0.030	3.569	0.3534
1024	4532.59	1053.34	0.232	3.568	0.3532

At $N = 1024$: current-relative errors.

Component	ROM %	FOM %
Displacement	3.568	0.353
Velocity	7.813	0.790
Energy state	6.214	0.608

Interpretation

The ROM is slower at every mesh. Displacement meets the descriptive target, but velocity and energy state do not.

Reference: the exact modal solution of the same spatial discretization. These are not continuum error bounds. Full host-to-host wave timing was not measured.

The FOM algorithm and tolerance change the conclusion

At $N = 1024$: all entries below are FOM/ROM runtime ratios.

Problem	Tight iterative	Fastest passing	Passing + transfers
Poisson 2D	75.45	33.91	14.90
Heat 2D	17.25	4.85	2.27
Burgers 2D	14.53	1.63	1.44
Reflective waves	0.23	0.23	Not measured

Direct sine-transform diagnostics:

Problem	GPU ms	Error %
Poisson 2D	0.317	0.0007351
Heat 2D	1.236	0.0003501
Reflective waves	16.624	0

Interpretation

Specialized direct solvers are faster on these separable linear problems. Iterative-FOM wins do not establish a universal FOM advantage.

Loose controls are selected on development cases; they are not globally optimal classical solvers. Poisson still misses the ROM target; Burgers retains its stall qualification. Wave direct displacement error is zero against the same-grid modal reference.

Reflective-wave cost is in latent evolution

At $N = 1024$:

Component	GPU ms
Initialization and projection	20.18
Latent evolution	4510.05
Dense GPU output	2.51

Takeaway

99.50% of query time is spent in latent evolution.

$T = 2.4$, $\Delta t = 0.0025$:

$960 \text{ steps} \times 4 = 3840 \text{ RK stages.}$

At every stage:

- Decoder Jacobian and directional curvature.
- QR factorization and triangular solve.
- SVD check of tangent rank.

Interpretation

The stage loop is fully reduced, but repeats expensive state-dependent work. Individual derivative, QR, SVD and GPU execution costs have not yet been separated.

Component medians need not sum exactly to the total median. One additional initial acceleration/rank check precedes the RK stages. Proposed next step: profile stage operations, then test cheaper guarding and fewer stages with accuracy checks.

What we can show now; what needs improvement

Supported by this development study

- Fixed learned weights transfer across the tested meshes.
- Reduced query time is nearly flat over this range.
- Heat beats the tested iterative FOM frontier on the finest mesh.
- Burgers has a smaller fine-grid gain with an explicit stopping limitation.

Next experiments, not current results

- Poisson: reduce physical error and diagnose representation versus projection limits.
- Burgers: measure stationarity and improve coarse-grid accuracy.
- Reflective waves: reduce stage cost; improve velocity and energy accuracy.
- Freeze choices, then evaluate independent final cases and training seeds.

A tunability claim requires an additional fixed-checkpoint physical-error versus cost study. Adjustable tolerances alone do not establish that tradeoff.

Proposed: tunability from one trained network

Train once per PDE, then select an inference configuration with unchanged weights.

Knob	Intended cost change	Current status
Active latent coordinates k	Smaller derivatives and solves	Subset accuracy unvalidated
Active spatial bank size R	Smaller operators and output map	Subset accuracy unvalidated
Solver budget / tolerance	Fewer or more corrections	Available with frozen weights
Time step Δt	Fewer or more evolution steps	Available for time-dependent PDEs

$$u_\ell(x) = \sum_{j=1}^{R_\ell} g_j(x) h_{\ell,j}(z_1, \dots, z_{k_\ell}).$$

Train shared weights with a loss at every supported level ℓ . At inference, choose a level and solver budget; physically omit inactive computation.

Proposed, not measured: screen truncation of current checkpoints as an ablation; train shared capacity levels if it is unreliable. Zero masks alone may not save runtime.

Proposed: demonstrate a physical accuracy–cost curve

First: current frozen checkpoints

- Sweep solve budgets/tolerances at fixed capacity; time steps separately.
- Pair physical error and full-query time; retain stopping failures.
- Find where additional computation stops improving physical accuracy.

Then: shared capacity levels

- Train with losses at each (k_ℓ, R_ℓ) ; preassemble each level's operators.
- Keep more weak equations than latent coordinates; validate each budget.
- Freeze choices, then compare with passing FOMs on independent cases.

Interpretation

Lower weak residual need not mean lower physical error. A tolerance setting does not guarantee a requested accuracy, and iterations cannot remove every representation limit.

Proposed figure: physical error versus query time, one curve per checkpoint; retain failures and all configurations. Select useful operating points on validation cases. This has not been demonstrated by the mesh-scaling tables.

Audit trail and development limitations

Panel	Job	GPU	Scientific source
Poisson 2D	3534456	A100 80 GB	085e318
Heat 2D	3529772	A100 40 GB	7e6d2e3
Burgers 2D	3534502	A100 40 GB	46ced2f
Reflective waves	3534457	A100 80 GB	e1d3779

GPU outliers, ascending mesh order:

Panel	ROM	Tight FOM
Poisson 2D	2/1/1	0/0/0
Heat 2D	0/0/0	0/0/0
Burgers 2D	0/0/0	0/0/0
Reflective waves	0/0/0	0/0/0

- GPU, float64 and highest precision verified; burn-in before timing.
- Paired full fields and repetition arrays retained; no outlier exclusions.
- Source/reference/field audits passed; archives checksum-verified.
- Final paper cases remain unopened.

Audit success does not imply physical accuracy or stationarity. Exact outlier rules and all controls are in the native audits and accompanying report/JSON.

Glossary: reading the tables and method

NMROM / FOM: nonlinear-manifold reduced model / solver on the full spatial grid.

N , mesh, intervals/axis: spatial cells along one coordinate; stored-node counts depend on boundaries.

GPU ms / transfers: online device time in milliseconds / movement of supplied inputs and requested outputs between CPU and GPU.

FOM/ROM: ratio of median times; above one means the ROM is faster for that named comparator.

Worst error: largest normalized field error across cases, output times and repetitions. Current-relative uses the reference norm at that time; fixed-initial uses its initial norm.

Target / refinement: physical-error threshold with numerical checks / disagreement after refining the reference or time step.

Frozen checkpoint / development cases: unchanged trained weights / already opened inputs used while developing the method.

Bank, G , R , head, z : learned spatial shapes, their grid matrix and count, nonlinear mixing map, and unknown latent coordinates.

Weak form / residual: averaging the PDE mismatch against smooth test functions / the mismatch in the solved equations.

CG / BiCGStab / Newton: iterative linear solvers / successive linearized corrections to nonlinear equations.

CN / midpoint / RK4: Crank–Nicolson / implicit midpoint / fourth-order Runge–Kutta time integration.

Stationarity / stall: small objective gradient / small progress or step, which need not imply stationarity.

Glossary: operators, numerical work and evidence

Preassembly / cached algebra: forming small operators before each online query and reusing them. Dense input/output work is still mesh-dependent.

Quadrature / sampled upwind: weighted spatial samples / evaluating the flow-direction-dependent difference operator at those samples.

Tensor: a multidimensional coefficient array encoding an operator; current Burgers is not the historical polynomial-tensor implementation.

FFT / sine transform / modal solution: transform algorithms that diagonalize these separable linear problems.

Preconditioning / Cholesky: making an iterative system easier to solve / a symmetric positive-definite matrix factorization.

Jacobian / curvature: how decoder output changes with latent coordinates / its directional second derivative.

QR / SVD / tangent rank: matrix factorizations used for least squares and singular values / the number of independent decoder directions.

Energy-state error: wave error measured jointly through velocity and displacement gradients; it is not merely the difference in total scalar energy.

Median / repetition / outlier: middle timing value / rerunning one input / an unusually slow repetition under the recorded rule. Repetitions are not independent physical cases.

Tunability / independent final validation: choosing inference effort at fixed weights / testing frozen choices on previously unopened cases.

Nested levels / masks / nondominated points: shared weights at several capacities / inactive terms / settings unbeaten in both cost and error.